

Name: _____ Date: _____

Period: _____

Problems 1–61. Solve each equation for all solutions (general solution). Express answers in exact form using π .

(a) $\sin \theta = \frac{\sqrt{2}}{2}$

$\theta = \frac{\pi}{4} + 2\pi k \quad \text{and} \quad \frac{3\pi}{4} + 2\pi k, \text{ for integer } k$

(b) $\cos \theta = -\frac{1}{2}$

$\theta = \frac{2\pi}{3} + 2\pi k \quad \text{and} \quad \frac{4\pi}{3} + 2\pi k, \text{ for integer } k$

(c) $\tan \theta = -1$

$\theta = -\frac{\pi}{4} + \pi k, \text{ for integer } k$

2. Solve $2 \sin \theta - 1 = 0$ for all θ in $[0, 2\pi]$. Show each step.Step 1 — Isolate $\sin \theta$: $\sin \theta = \frac{1}{2}$ Step 2 — Reference angle: $\arcsin\left(\frac{1}{2}\right) = \frac{\pi}{6}$ Step 3 — Identify quadrants and write solutions: **Sine is positive in Q1 and Q2.** $\theta = \frac{\pi}{6}$ and $\theta = \pi - \frac{\pi}{6} = \frac{5\pi}{6}$.3. Solve $2 \cos^2 \theta - \cos \theta - 1 = 0$ for all θ in $[0, 2\pi]$.Let $u = \cos \theta$. Factor: $(2u + 1)(u - 1) = 0$

Case 1: $u = -\frac{1}{2} \Rightarrow \theta = \frac{2\pi}{3}$ and $\frac{4\pi}{3}$

Case 2: $u = 1 \Rightarrow \theta = 0$ (and 2π , same point)

Teacher NoteFull solution set in $[0, 2\pi]$: $\theta \in \{0, \frac{2\pi}{3}, \frac{4\pi}{3}\}$. Students often miss $\theta = 0$ because $\arccos(1) = 0$ is on a quadrantal angle.

4. $h(t) = 8 \sin\left(\frac{\pi}{15}t\right) + 10$, $0 \leq t \leq 60$.

(a) For what values of t in $[0, 60]$ is $h(t) = 14$?**Set** $8 \sin\left(\frac{\pi}{15}t\right) + 10 = 14$. **Then** $\sin\left(\frac{\pi}{15}t\right) = \frac{1}{2}$. **Reference angle:** $\frac{\pi}{6}$. **In** $[0, 2\pi]$: $\frac{\pi}{15}t = \frac{\pi}{6} \Rightarrow t = 2.5$ s; and $\frac{\pi}{15}t = \frac{5\pi}{6} \Rightarrow t = 12.5$ s. **Adding one period** ($T = 30$): $t = 32.5$ and $t = 42.5$ s. **All four are in** $[0, 60]$.(b) For what values of t in $[0, 60]$ is $h(t) \geq 14$?

$[2.5, 12.5] \cup [32.5, 42.5]$

5. (AP-Style MC)

(B) $\theta = \frac{\pi}{3}$ and $\theta = \frac{5\pi}{3}$ ← correct

- (A) misses the Q4 solution.

- (C) gives the general form, not just $[0, 2\pi]$.
- (D) and (E) are the solutions for $\sin \theta = 1/2$ and $\cos \theta = -1/2$, respectively.

6. (AP-Style MC)

(E) Both (A) and (B) ← **correct**

- Both $\sin \theta = 0.6$ and $\cos \theta = 0.6$ have exactly two solutions in $(0, 2\pi)$ (one in Q1, one in Q2 for sine; one in Q1, one in Q4 for cosine).
- Tangent is one-to-one per period so has only one solution per period.

Extension optional

Extension (Optional): $2 \sin^2 \theta - \sin \theta - 1 = 0$

(a) $(2 \sin \theta + 1)(\sin \theta - 1) = 0$

(b) Case 1: $\sin \theta = -\frac{1}{2}$: $\theta = \frac{7\pi}{6} + 2\pi k$ **and** $\theta = \frac{11\pi}{6} + 2\pi k$

Case 2: $\sin \theta = 1$: $\theta = \frac{\pi}{2} + 2\pi k$

(c) In $[0, 4\pi]$: $\frac{\pi}{2}, \frac{7\pi}{6}, \frac{11\pi}{6}, \frac{5\pi}{2}, \frac{19\pi}{6}, \frac{23\pi}{6}$ — **six solutions total.**

Preview — Lesson 3.11: Secant, Cosecant, and Cotangent

You have now solved equations involving sine, cosine, and tangent. In Lesson 3.11 we meet three reciprocal functions — secant, cosecant, and cotangent — and explore how their graphs and asymptotic behavior arise directly from the functions you already know.